Vaibhav Raheja

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PROFESSIONAL SUMMARY

Robotics Systems Engineer with 2+ years building hardware-software interfaces and deploying autonomous platforms on embedded compute. Experienced integrating sensor drivers and ROS2 layers for LiDAR, cameras, and embedded devices on NVIDIA Jetson. Demonstrated results: 90% fleet uptime, 40% improvement in hardware fault detection, and production deployments across multi-sensor stacks in safety-critical field environments.

WORK EXPERIENCE

TerraWise Solutions, Inc, *Founding Engineer* — Mar 2026 – Present | San Francisco, CA

- Leading platform bring-up on NVIDIA Jetson: integrating ODrive motor controllers via CAN bus and establishing ROS2 driver interfaces from scratch.
- Defining hardware-software integration standards and Docker deployment workflows, enabling reproducible embedded deployments across robot units.
- Implementing LiDAR-based stop-by-obstacle detection as part of the autonomous navigation safety layer.

EarthSense, Inc, *Robotics Engineer / Project Lead* — Aug 2024 – Mar 2026 | Champaign, USA

- Authored driver interfaces integrating Hesai LiDAR, PTZ cameras, and Seek thermal sensors into the ROS2 production stack across 3+ multi-site robot units.
- Built real-time hardware health monitoring and telemetry pipelines, achieving 40% improvement in fault detection rates and 90% fleet uptime.
- Shipped 35+ PRs across the motion control and autonomy stack in C++ and Python; stress-tested hardware at 40°C+ across multiple solar farm field deployments.
- Enabled low-latency teleoperation via an AWS Kinesis WebRTC proof of concept for remote fleet monitoring.
- Led a cross-functional team of 8+ engineers delivering an autonomous solar panel inspection platform across production solar farm deployments.

Intelligent Motion Laboratory, *Robotics Research Developer* — Aug 2023 – Dec 2023 | Champaign, USA

- Achieved sub-2mm tracking accuracy for autonomous robotic eye examinations using MediaPipe FaceMesh and ZED depth cameras on UR5.
- Designed a custom camera mount in Fusion 360 for the UR5, improving head tracking precision by 20%.

All India Institute of Medical Sciences (AIIMS) Hospital, *Robotics Research Developer, NMIMS* — Feb 2021 – May 2023 | Mumbai, India

- Engineered a robot-assisted intubation system on xArm5 with HOTAS teleoperation and real-time camera feedback for critical care.
- Fabricated a custom catheter end-effector with integrated miniature camera for real-time airway visualization.

SKILLS

Programming: Python, C++, ROS/ROS2, OpenCV, PyTorch, Machine Learning (ML), CNNs
Robotics & Perception: LiDAR Integration (Hesai), Multi-Sensor Fusion, Camera Calibration, Path Planning, Motion Planning, Control Algorithms, Reinforcement Learning, Gazebo, SLAM, Docker
Tools: Linux, Git, Fusion 360, CAD, Arduino, Raspberry Pi, 3D Printing

PROJECTS

Benchmarking Control Algorithms for Unitree Go1 Robot — *Python, ISAAC Sim, Reinforcement Learning* | *2024*

- Built an ISAAC Sim benchmarking framework for the Unitree Go1, comparing factory vs. RL-based locomotion across velocity tracking, stability, and terrain adaptability.
- Verified real-world RL controller performance by deploying "Walk These Ways" on a physical Go1 across grass, gravel, and inclines.

GRAIC Autonomous Racing — *Python, Hybrid A*, Path Planning, PD Control, Vehicle Dynamics* | *2023*

- Achieved 40.8% lap time improvement on the Shanghai circuit using a Hybrid A* path planner tuned to vehicle kinematics and non-holonomic dynamics.
- Benchmarked three algorithms (Hybrid A*, Dynamic Programming, Spline Interpolation) and tuned a PD controller for steering and velocity.

Intelligent Ground Vehicle Competition (IGVC), Team DARVIN — *Python, ROS, OpenCV, ODrive, Velodyne LiDAR, Gazebo* | *2019–2023*

- Served as Co-Captain, securing 2nd in the Cyber Challenge (2023) and 3rd in the Auto-Nav Challenge (2022) against 15+ international university teams.
- Developed SOCRATES 2.0 on ROS with Velodyne LiDAR avoidance, OpenCV lane detection, GPS navigation, and ODrive S1 motor control for the competition.

EDUCATION

University of Illinois Urbana-Champaign — Aug 2023 – Dec 2024 | Champaign, USA
Master of Engineering in Autonomy and Robotics

Mukesh Patel School of Technology Management & Engineering — Jul 2019 – Jun 2023 | Mumbai, India
Bachelor of Technology in Computer Engineering

PUBLICATIONS

V. Raheja, V. Shah, M. Shetty, and P. Patel, "Multi-Disease Prediction System using Machine Learning," *2022 International Conference on Futuristic Technologies (INCOFT)*, Belgaum, India, 2022, pp. 1–6. doi: 10.1109/INCOFT55651.2022.10094382